

Online industrial fault prognosis in dynamic environments via task-free continual learning

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ARTICLE INFO

Communicated by D. Wu

Keywords:

Fault prognosis
Online learning
Continual learning
Mixture of experts
Bayesian nonparametric framework

ABSTRACT

Online industrial fault prognosis based on real-time condition monitoring plays a significant role in the predictive maintenance of the interested system in service. Despite the recent advances in deep learning-based fault prognosis, most existing works focus on offline tasks using static deep models, which cannot flexibly adapt their learning behavior to the real-time health conditions of the interested machines. To meet more realistic scenarios, it is essential but challenging to develop an online prognostic method that is capable of incrementally learning and flexibly adapting to different tasks in dynamic environments. Meanwhile, continual learning as a new learning paradigm aims at learning adaptively like humans. This paper proposes a novel online fault prognosis paradigm via task-free continual learning towards more practical industrial scenarios. Correspondingly, an intelligent fault prognosis method based on continual neural Dirichlet process mixture is developed for this problem. It dynamically allocates new resources to learn new tasks by integrating the mixture of experts with the Bayesian nonparametric framework, wherein each expert with two components is designed to perform remaining useful life prediction and task identification respectively. The comprehensive experimental results on two turbofan engine datasets validate the proposed method, which offers a promising solution for online fault prognosis under more open settings.

1. Introduction

Industrial fault prognosis aims at increasing the availability and safety of the engineering system, while reducing maintenance cost based on condition monitoring (CM) and the remaining useful life (RUL) prediction, which plays a critical role in predictive maintenance [1]. In recent years, the rapid development of the Industrial Internet of Things (IIoT) has boosted prognostics and health management (PHM). In particular, deep learning has been attracting increasing attention due to its ability to automatically capture degradation features from raw CM data without relying on too much prior knowledge of experts [2]. In the existing literature, many methods based on deep learning have been successfully developed for fault prognosis [3–7].

Though these studies achieved impressive prognostic performance, they mostly focus on offline scenarios with ideal environments, i.e. the invariant operating condition and the independent and identical (i.i.d.) CM data stream, where the deep models were trained using CM data that were shuffled to maintain i.i.d. assumption and remarkable performance improvement was obtained by revisiting the CM data over multiple epochs. Based on the historical CM data, the parameters of the offline models were statically estimated and were not adaptive to the interested system in service. However, a complex engineering

system in real industries generally experiences different degradation patterns under varying operating conditions throughout its lifetime. This results in quite dynamic and nonlinear degradation process with a non-stationary data stream, where the general distribution of the CM data can change rapidly with the progress of monitoring. Moreover, for safety and reliability reasons, the key industrial equipment in service typically has a long life and therefore requires long-term online health monitoring, which calls for online learning for real-time fault prognosis. Predicting its remaining useful life across different operating conditions for each machine is akin to handling distinct tasks. However, the challenge lies in the complete uncertainty about the number, types, and transition points of these tasks while the machine is still in operation. This ambiguity adds a layer of complexity to forecasting and underscores the practical challenges in real-time fault prognosis. Hence, it is quite essential to design deep models that can continually adapt to the non-stationary data stream and keep capturing the degradation process over time for online fault prognosis. However, general deep neural networks suffer from the catastrophic forgetting issue when sequentially learning multiple tasks one-by-one due to the interference with previously learned knowledge as the models are trained with new information [8]. Continual learning (CL) which aims

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<https://doi.org/10.1016/j.neucom.2024.127930>

Received 22 May 2023; Received in revised form 1 April 2024; Accepted 27 May 2024

Available online 29 May 2024

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at learning from an infinite non-iid stream of data in the similar way to humans without catastrophically forgetting the knowledge learned from previous tasks [9], holds great potential to solve this challenging problem.

However, most existing continual learning approaches assume that the data stream is split into a sequence of different tasks that are explicitly known during the training process, which is far from realistic since the boundaries among many tasks are not distinguishable in many real scenarios. Task-free CL is a general CL that does not require a specific task definition nor clear task boundaries under the gradual drift of data distribution. Thus, it is demanding and holds great potential to put the fault prognosis of complex engineering machines towards more intelligent and more practical in real industries, but has been largely understudied. Meanwhile, existing continual learning can be classified into three categories: replay, regularization and expansion-based methods. Particularly, the expansion-based methods can eliminate catastrophic forgetting via avoiding pre-existing components from being overwritten by the newly received information [10]. This paper aims to study a more practical industrial fault prognosis with an open set, wherein several challenging issues are considered: (1) long-term online learning based on the non-stationary CM data stream for real-time RUL prediction; (2) complex CM data distribution due to the varying operating conditions and different fault modes; (3) continual learning that avoids catastrophic forgetting while learning without clear task boundaries over the CM data stream.

To effectively address these challenges, this work proposes a task-free continual learning-based approach for online fault prognosis of complex engineering systems, where an expansion-based method is adopted for task-free CL. Enlightened by the recently proposed Continual Neural Dirichlet Process Mixture (CN-DPM) model [10] that comprises a mixture of experts responsible for different clusters of the data stream, the proposed method extends the CN-DPM model to general online fault prognosis tasks, where the exact boundaries of RUL prediction tasks and task descriptions are unavailable at both training and test time. Based on the Bayesian nonparametric framework, the task-free fault prognosis is regarded as learning of a Dirichlet process mixture (DPM) model with neural experts, the number of which is not prefixed and can adaptively grow based on the non-i.i.d. stream of CM data. The main contributions of this work are highlighted as follows.

1. This work studies a new fault prognostic paradigm named online task-free continual fault prognosis (OTFCFP) that aims to sequentially perform multiple real-time fault prognosis tasks under dynamic nonstationary CM data stream, where the task definition and boundaries are unavailable. The OTFCFP problem with more open settings tends to be more general and more practical in real industries.
2. A novel deep model based on CN-DPM is developed to solve the OTFCFP problem, in which a mixture of experts is integrated with the Bayesian nonparametric framework to eliminate catastrophic forgetting while accumulating knowledge of previous tasks for forward transfer. Pretrained expert models can also be used to incorporate knowledge of historical data.
3. A regression model and a generative model are combined to build an expert, wherein the temporal convolutional network (TCN) attached with attention model is used as the regression model to enhance the temporal feature representations and efficiently perform RUL prediction, while the generative model combines VAE with GRU network to identify the task boundaries under the task-free setting.

2. Related works

In real industrial scenarios, it is quite common that the complex engineering systems work under time-varying operating conditions or have different fault modes, which brings a distribution shift of the

CM data, decreasing the generalization ability of the deep models. Deep transfer learning techniques, such as domain adaptation [11] and adversarial learning [12], have been developed to address this issue, which exploits the existing knowledge of another task or domain to aid the learning process of a given task. Recently, Zhang et al. [13] proposed a transfer learning method for cross-domain fault prognosis problems, where the training and testing CM data are from different system operating conditions. This method uses deep representation regularization to perform data alignment in high-level feature space and promotes the prognostic knowledge transfer across different operating conditions. A deep adversarial networks method is proposed by Li et al. [14] to cope with potential sensor malfunctions in real industrial scenarios. The method incorporates a global feature extraction scheme to comprehensively exploit information from diverse sensor sources. Through a systematic exploration of both global and shared features, the proposed approach substantiates its efficacy, manifesting commendable and robust predictive performance in estimating RUL under testing scenarios marked by sensor malfunctions.

Despite the promising advances of deep transfer learning in industrial fault prognosis, it only relaxes the classical machine learning assumption between two different domains, and it does not involve any continuous adaptation and backward transfer after learning the target task [15]. A more practical and general machine learning paradigm requires online learning from a continuous stream of data without catastrophic forgetting, rather than offline training under the setting of large batches or separate tasks. Continual learning shifts this paradigm towards deep neural networks that are capable of continually learning from nonstationary data stream and accumulating knowledge over a sequence of different tasks without retraining from scratch. In the past years, continual learning has been attracting growing interest in both academic and industrial fields. To develop effective continually adapting systems, the main obstacle that needs to be overcome is the catastrophic forgetting of previously acquired knowledge as new information is learned [16]. Humans can effectively retain part of the previously learned information while learning new concepts sequentially, which is an example of such systems [17]. As a more realistic but challenging learning paradigm, online task-free continual learning under a more general setting that the non-i.i.d. data stream is learned online without explicit task boundary information, has been largely understudied. Recently, Lee et al. [10] proposed a deep model named CN-DPM for task-free CL, which employs a set of neural network experts to process different subsets of the data stream and the number of experts can be adaptively expanded based on rules under the Bayesian nonparametric framework. The experimental results on image classification and generation verify its effectiveness.

In the field of industrial fault prognosis, there are few literatures on the continual learning. Maschler et al. [18] presented a continual learning-based approach for fault prediction of turbofan engines using LSTM associated with elastic weight consolidation, which gives a try to use regularization-based method to constrain the change of important parameters in the previously learned tasks. The experiment results show that it works when different tasks are related but its performance deteriorates when the tasks are of vastly different complexity. Moreover, this work requires clear task boundaries and learns the model on each task in an offline way, which is limited in more practical scenarios. To the best of our knowledge, this paper is one of the first to study an online task-free continual learning-based approach for industrial fault prognosis, aiming to offer an effective solution for this open and practical issue in real industries.

3. Proposed approach

3.1. Problem setting

This work investigates the online task-free continual industrial fault prognosis based on real-time CM data, in which the collected data

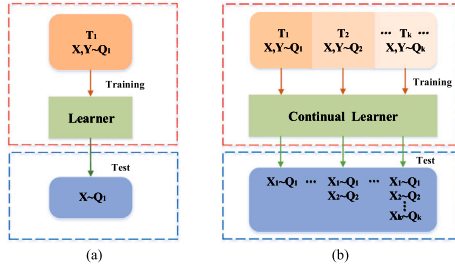


Fig. 1. The comparison of different learning paradigms: (a) standard supervised learning; (b) general continual learning.

samples with multi-sensor and operating condition information arrive sequentially. Typically, this study will be carried out under these settings: (1) purely online learning that learns from a continuous data stream without large batches or multiple epochs that are used in offline learning. (2) time-varying operating conditions and different fault modes exist, leading to a non-stationary CM data stream over time. (3) there is neither an explicit task definition nor clear task boundaries among different tasks.

Fig. 1 shows the comparison between standard supervised learning and general CL. Different from the standard machine learning that all the training data need to be usable before learning under i.i.d. assumption, the goal of CL is to learn a model f parameterized by θ that optimizes a predefined loss ℓ on the new samples (x^t, y^t) without forgetting previously learned knowledge and possibly improving the performance on those tasks [19].

$$\begin{aligned} \theta^t &= \underset{\theta, \xi}{\operatorname{argmin}} \ell(f(x^t; \theta), y^t) + \sum \xi_i \\ \text{s.t. } \ell(f(x^i; \theta), y^i) &\leq \ell(f(x^i; \theta^{t-1}), y^i) + \xi_i, \\ \xi_i &\geq 0, \forall i \in [0, \dots, t-1]. \end{aligned} \quad (1)$$

where $\xi = \{\xi_i\}$ denotes a vector of slack variables that allows a small increase in the losses of previous samples.

The CL scenario in fault prognosis can be formulated as CM data flow containing various tasks $T_1, T_2, \dots$, where each task T_n includes a subset drawn from a certain data distribution $p(x|z=n)$. As the fault prognosis tasks are essentially discriminative tasks, the overall distribution can be expressed as the compound of K conditional distributions:

$$p(y|x) = \sum_{n=1}^K p(y|x, z=n) p(z=n|x). \quad (2)$$

Since z is unknown in task-free CL, the posterior distribution $p(z|x, y)$ should be inferred by the model itself. Considering a certain time step t in an infinite CM data stream, the system collects a small batch of samples $\{(x^i, y^i)\}_{i=1}^{N_b}$ sampled non-i.i.d. from a dynamic data distribution Q which can suddenly or gradually change over time, where N_b denotes the number of the certain time steps t of a batch.

Based on the above learning paradigm, this paper aims to develop an intelligent online task-free continual fault prognosis model, which continually learns knowledge from the online non-i.i.d. CM data stream, and can adapt the non-stationary data distribution by automatically identifying the unavailable task boundaries. Besides, it does not require storing large amounts of data for training multiple epochs, which is more practical but challenging in real scenarios.

3.2. Continual neural Dirichlet process mixture (CN-DPM)

To model the overall conditional distribution in Eq. (2), expansion-based CL methods preserve the knowledge of previous tasks with a mixture model by isolating the model parameters of different tasks, i.e., learning new task distribution with new parameters and leaving the existing parameters intact. A key problem that arises in expansion methods is how to choose the proper expert given x , i.e. inferring $p(z|x)$.

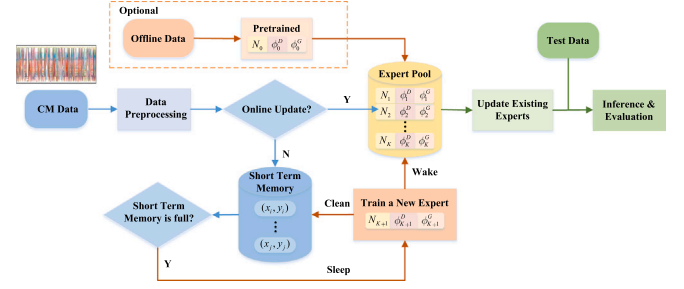


Fig. 2. The overview of the online task-free continual fault prognosis via CN-DPM model.

Since explicit task descriptor z is unknown, a gating mechanism is used to infer $p(z|x)$ and decide which expert should be used to process x , making the RUL prediction task naturally reduce to the mixture of experts (MoE) formulation that performs prediction with the total of all the expert outputs with their corresponding gate values as the weight,

$$p(y|x) \approx \sum_n p(y|x; \phi_n) p(z=n|x). \quad (3)$$

Nevertheless, the gate network is a classifier that finds proper experts for given samples, and a single gate network would suffer from catastrophic forgetting issue if trained online.

To address this issue in general task-free CL, the newly developed CN-DPM model builds a gating mechanism, which can overcome forgetting, i.e.,

$$\begin{aligned} p(y|x) &\approx \sum_n p(y|x; \phi_n^D) p(z=n|x) \\ &\approx \sum_n p(y|x; \phi_n^D) \frac{p(x; \phi_n^G) p(z=n)}{\sum_{n'} p(x; \phi_{n'}^G) p(z=n')}, \end{aligned} \quad (4)$$

where the superscript D and G denote the discriminant model for regression and generation model for gating respectively.

Even the total number of experts is also unknown beforehand, therefore, Dirichlet process mixture (DPM) is employed in the CN-DPM model to estimate the mixed distribution that has an unknown quantity of ingredients. Since it is infeasible to infer the exact posterior of DPM models, sequential variational approximation (SVA) [20] is adopted to approach the posterior online. Overall, the CN-DPM model is made up of a group of experts. Every expert applies a discriminant model $p(y|x, z=n)$ to predict RUL, while a generative model is used to estimate the marginal likelihood $p(x|z=n)$. Specifically, the regressor of the n th expert is expressed as $p(y|x; \phi_n^D)$, and $p(x; \phi_n^G)$ represents its density estimator, in which ϕ_n^D and ϕ_n^G represent the model parameters. Thus, the predicted RULs $p(y|x)$ could be derived from Eq. (4) via merging the output of the regressor and the generative model. The overview of the online task-free continual fault prognosis based on CN-DPM model is presented in Fig. 2. When a newly arriving sample is available, it is first preprocessed and then judged whether it can be covered by the pre-existing experts. If the new sample can be covered, an online update is adopted to iterate expert pool, otherwise the new sample is stored in the STM. While storing each sample, the storage of STM is checked. When the STM is full, the model suspends receiving new CM data and trains a new expert using data from STM. After several epochs, the number of experts is expanded and the online data inflow recovers.

3.3. Expert design in CN-DPM

As mentioned beforehand, every expert in CN-DPM is associated with a discriminant model to model $p(y|x; \phi_n^D)$ and a generation model to model $p(x; \phi_n^G)$, jointly fitting $p(x, y; \phi_n)$. In this work, the discriminative component adopts attention-based temporal convolutional

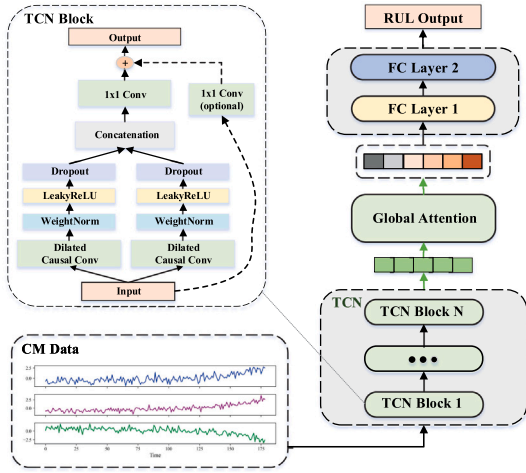


Fig. 3. Attention-based TCN for RUL prediction, which is used as the discriminative model of a certain expert in the CN-DPM framework.

networks (TCN) to perform RUL prediction as a regression model, while a GRU-based variational auto-encoder (VAE) is used as a density estimator.

3.3.1. Attention-based TCN as discriminative model

In real industries, CM data acquired from various sensors are generally time series data and the RUL prediction task can be essentially regarded as a time series regression problem. TCN serves as a powerful deep model that distills some superior operators of convolutional neural networks into an easy but efficient structure. It outperforms general recurrent structures in different sequence modeling tasks [21,22]. As shown in the upper left of Fig. 3, a TCN block consists of two parallel dilated causal convolutional layers with non-linearity from LeakyReLU. Weight normalization facilitates the training procedure and makes the architecture perform better. A spatial dropout is attached to follow every convolution layer as a regularization term. Besides, an optional 1×1 convolution is adopted to adapt the width discrepancy between the input and output. Please refer to [21] for further understanding of the TCNs.

Furthermore, the attention mechanism has the potential to capture the dependencies that play an important role in the task, which contributes auxiliary information to the distance-aware representations modeled by the deep neural networks. In this study, TCN is equipped with an attention model to enhance the feature learning in RUL prediction, as shown in Fig. 3. In the global attention module, the examples from raw CM data are utilized as query vectors, which are paired with the feature embeddings output by the TCN blocks to calculate the alignment scores. This facilitates bridging the relation between the sensor variables and the feature representations and enhances the performance of RUL prediction. We train the attention-based TCN model with the following objective function:

$$L_y = \frac{1}{N_s} \sum_{(y_s^i, x_s^i) \in D_s} L_y(G_y(G_a(h_o^i, h_s^i)), y_s^i), \quad (5)$$

where G_y denotes the regression model for RUL prediction, G_a denotes the attention module, h_o^i and h_s^i denote high-level features of operating data and sensor data respectively and y_s^i denotes the label of the data sample. Mean square error is adopted as the regression error metric and the main objective is to minimize the prediction error.

3.3.2. GRU-based VAE as generative model

Besides the discriminative model for RUL prediction, the expert still requires a density estimator to describe the marginal likelihood $p(\mathbf{x}|z = n)$. VAEs have been shown to be an effective deep generative

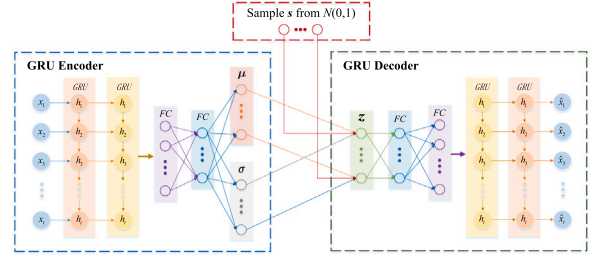


Fig. 4. The overview of the generative model, which is comprised of GRU-based encoder and decoder.

model that recovers complex multimodal distributions over the data space [23]. Typically, VAE parameterizes the likelihood with a highly flexible function approximator such as a deep neural network denoted as $p(\mathbf{x}; \phi_n^G)$, which uses a variational approximation $q(z|\mathbf{x})$ of the posterior that enables the usage of the evidence lower bound:

$$\mathcal{L}_{\text{vae}} = -D_{KL}(q_\phi(z|\mathbf{x})||p_\theta(z)) + \mathbb{E}_{q_\phi(z|\mathbf{x})}[\log p_\theta(\mathbf{x}|z)], \quad (6)$$

where ϕ and θ are the parameters of the encoder and decoder, respectively. D_{KL} is the Kullback–Leibler divergence between the prior $p_\theta(z)$ and the variational approximation $q_\phi(z|\mathbf{x})$ of $p(z|\mathbf{x})$. $\mathbb{E}_{q_\phi(z|\mathbf{x})}$ is the reconstruct loss reflecting the difference between the VAE-generated result and the input. The VAE optimizes the parameters of the networks by maximizing the lower bound of the log likelihood $\mathcal{L}_{\text{vae}}$.

Since the CM data are comprised of multi-sensor time series signals, we introduce the temporal dependency of time-series data in the VAE, by integrating the GRU networks into the VAE. Fig. 4 shows the structure of the GRU-based VAE. Given an input $\mathbf{x}_t$ at time t , the GRU-based encoder models the posterior $p(z|\mathbf{x})$ with two linear layers to estimate the mean and covariance.

3.4. Online model optimization

Based on the CN-DPM model, this work develops an intelligent fault prognosis approach to perform online task-free continual learning in a practical way. The training of the whole model mainly consists of two steps: (1) adaptively decide on the growth of the experts, i.e. the exact number of the experts; (2) optimize a certain expert consisting of attention-based TCN for RUL prediction and GRU-based VAE for density estimation.

To this aim, the data samples are supposed to arrive one-by-one sequentially during the training process, and each arrived sample would be firstly assessed whether it can be covered by the pre-existing experts or another expert is supposed to be newly introduced. Concretely, assume a sequence of samples including $(\mathbf{x}_t, y_t)$ have been handled and K experts have been also introduced when a newly arriving sample $(\mathbf{x}_{t+1}, y_{t+1})$ is available. The responsibility $r_{t+1,n}$ can be derived as below:

$$r_{t+1,n} \propto \begin{cases} (\sum_{i=1}^t r_{i,n}) p(y_{t+1} | \mathbf{x}_{t+1}; \hat{\phi}_n^D) p(\mathbf{x}_{t+1}; \hat{\phi}_n^G), & \text{if } 1 \leq n \leq K; \\ c p(y_{t+1} | \mathbf{x}_{t+1}; \hat{\phi}_0^D) p(\mathbf{x}_{t+1}; \hat{\phi}_0^G), & \text{if } n = K + 1, \end{cases} \quad (7)$$

wherein $\hat{\phi}_0 \sim G_0(\phi)$ and G_0 denotes a distribution that corresponds to the parameter initialization of the neural networks. It is worth mentioning that if a certain amount of historical CM data is available, it is optional to pre-train an offline expert, then its weights can be endowed as G_0 to finetune the subsequent online experts. In this way, some knowledge from the offline CM data can be transferred to enhance the online learning. If $r_{t+1,K+1}$ is not the largest one, this sample will be sent to the available experts in proportion to $r_{t+1,n}$, then the gradient descent (GD) algorithm with the learning rate η is adopted to obtain the

maximum posterior estimation $\hat{\phi}$ given the new sample assigned up to the current time step:

$$\hat{\phi}_n^{(t+1)} \leftarrow \hat{\phi}_n^{(t)} + \eta \left(\nabla_{\hat{\phi}_n^{(t)}} \log G_0(\hat{\phi}_n^{(t)}) + \nabla_{\hat{\phi}_n^{(t)}} \log p(\mathbf{x} | \hat{\phi}_n^{(t)}) \right). \quad (8)$$

Otherwise, a new expert will be created for the new sample. Since both the two components in an expert are deep neural networks, using only a single sample to train the new expert at each time step results in serious overfitting. To alleviate the problem, a short-term memory (STM) with maximum capacity S is employed to store the CM data samples that are assessed as new, and the online data inflow will be stopped temporarily when the STM is full. Then a new expert will be expanded and trained over multiple epochs until convergence with these data from the STM. The overall training procedure is presented in Algorithm 1, where $r_{t,K+1}$ is replaced by $r_{t,0}$ for brevity. In addition, mean square error (MSE) is used as the regression loss function to update the model parameters of the attention-based TCN (ATCN), and Adam optimizer is employed for online model optimization.

Algorithm 1: Training procedure of the online task-free continual learning-based fault prognosis method

Require: CM data $(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_T, y_T)$, Concentration c , base measure G_0 , learning rate η , STM capacity S .

- 1 $S \leftarrow \emptyset$ {Initialize an empty STM};
- 2 $N_0 \leftarrow c$; $K \leftarrow 0$ {Start with zero experts};
- 3 **if** Historical CM data are available **then**
- 4 $\hat{\phi}_0 \leftarrow$ Pretrain the expert (E_0)
- 5 **else**
- 6 $\hat{\phi}_0 \leftarrow$ RandomSample (G_0)
- 7 **end**
- 8 **for** $t = 1$ **to** T **do**
- 9 **for** $n = 0$ **to** K **do**
- 10 $p(y_t | \mathbf{x}_t; \hat{\phi}_n^D) \leftarrow$ The regression model ATCN ;
- 11 $p(\mathbf{x}_t; \hat{\phi}_n^G) \leftarrow$ The generative model GRU-VAE ;
- 12 $a_n \leftarrow p(y_t | \mathbf{x}_t; \hat{\phi}_n^D) p(\mathbf{x}_t; \hat{\phi}_n^G)$; $r_{t,n} \leftarrow N_n a_n$;
- 13 **end**
- 14 $r_{t,0:K} \leftarrow r_{t,0:K} / \sum_{n=0}^K r_{t,n}$;
- 15 **if** $\arg \max_n r_{t,n} = 0$ **then**
- 16 $S \leftarrow \{(\mathbf{x}_t, y_t)\} \cup S$;
- 17 **if** $|S| \geq S$ **then**
- 18 $\hat{\phi}_{K+1} \leftarrow$ Train new expert (S, G_0);
- 19 $N_{K+1} \leftarrow |S|$; $S \leftarrow \emptyset$;
- 20 $K \leftarrow K + 1$;
- 21 **end**
- 22 **else**
- 23 $r_{t,1:K} \leftarrow r_{t,1:K} / \sum_{n=1}^K r_{t,n}$;
- 24 **for** $n = 1$ **to** K **do**
- 25 $N_n \leftarrow N_n + r_{t,n}$;
- 26 $\hat{\phi}_n \leftarrow \hat{\phi}_n + r_{t,n} \eta \nabla_{\hat{\phi}_n} \log a_n$;
- 27 **end**
- 28 **end**
- 29 **end**

4. Experimental study

In this section, two case studies concerning different prognostic tasks under online task-free CL setting are carried out to validate the effectiveness of our proposed method. The experimental results and some comparison analyses are presented.

4.1. Datasets overview

4.1.1. CMAPSS dataset

The CMAPSS dataset has been widely used for offline fault prognostic studies, and it comprises a fleet of turbofan engines with degradation

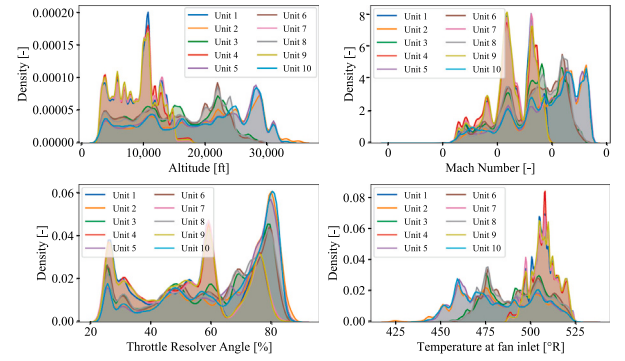


Fig. 5. Density distributions of the flight envelopes given by the operating conditions in DS01 of the N-CMAPSS dataset.

data produced by the Commercial Modular Aero-Propulsion System Simulation (C-MAPSS) testbed [24]. This dataset consists of 4 sub-datasets with diverse fault modes and operating conditions characterized by Mach number, sea-level temperature and altitudes, etc, leading to complex scenarios. Each sub-dataset has a certain number of engines, each of which includes a sequence of multivariate trajectories that are sampled from 21 sensor outputs over a period of operating cycles. In addition, those engines start from an unknown initial state of health condition and gradually exhibit nonlinear degradation following the time-varying operating conditions. Table 1 summarizes some basic information about the CMAPSS dataset. Since this paper focuses on online continual learning, four sub-datasets from FD001 to FD004 are used to form four different sequential tasks, where the number and boundaries of the tasks are unknown during the learning process.

4.1.2. N-CMAPSS dataset

The N-CMAPSS was recently developed as a new realistic dataset that contains a fleet of aircraft engines with run-to-failure trajectories. Due to the lack of significant complexity factors, the CMAPSS dataset has a fidelity gap between the simulated degradation trajectories and that in real engines. In contrast, real flight conditions recorded from an actual commercial plane are considered in N-CMAPSS, meanwhile, the degradation modeling is also extended by building the connection between the degradation process and its operation history [25]. Thus, higher fidelity is endowed to the N-CMAPSS dataset, which facilitates to the development of deep learning-based prognostic methods that can be easily transferred to real scenarios. Concretely, the N-CMAPSS dataset consists of eight sub-datasets, where two distinctive fault modes that affect the efficiency or flow of all the rotary sub-components are present. As seen in Table 1, four sub-datasets (DS01, DS02, DS04, DS05) are used to form four different tasks. Fig. 5 shows the operation distributions of the flight envelopes presented by multiple variables that describe different scenarios for dataset DS01. It can be clearly seen that there are distinct operation distributions among different units, corresponding to various flight profiles during the training process. More details about the dataset can be found in [25].

4.2. Experimental setup

For data preprocessing of the CMAPSS dataset, the actual RUL labels are described with a piece-wise linear degradation model in the training period as same as the work [5]. Regarding the N-CMAPSS dataset, interval sampling is adopted to collect the input data due to too long data sequences. Time windows are used to process and enhance the temporal information of the inputs.

In our study of both datasets, each subdataset is designated as an individual task and assigned and labeled for clarity. Notably, variations exist among tasks concerning the number of engines included in the

Table 1
Basic information of the CMAPSS dataset and the N-CMAPSS dataset.

Dataset	Subdataset	Engines in train/test set	Fault mode	Operational condition	Task ID
CMAPSS	FD001	100/100	1	1	1
	FD002	260/259	1	6	2
	FD003	100/100	2	1	3
	FD004	249/248	2	6	4
N-CMAPSS	DS01	6/4	1 (HPT)	N/A	1
	DS02	6/3	2 (HPT, LPT)	N/A	2
	DS04	6/4	1 (Fan)	N/A	3
	DS05	6/4	1 (HPC)	N/A	4

train and the test set. Furthermore, distinct differences emerge in the count of fault modes associated with each task. The detailed allocation of tasks, along with their respective characteristics, is outlined in Table 1, presenting a comprehensive view of the experimental setup for the datasets.

Note that the online continual learning paradigm defines a sequence of different tasks where these streams of CM data are assumed to be non-stationary, and the task information is unavailable (task-free) to the prognostic model at both the training and test stage. Besides, to enable a completely online setting, the data of every task is only displayed for one epoch with a small batch size of 16. For the performance evaluation of RUL prediction, root mean squared error (RMSE) and score function [24] are used for both two cases. Moreover, inspired by the forgetting measure [26] that estimates how much the model forgot the knowledge of previous tasks given its current state, we define an average forgetting measure AF_k to quantify the forgetting at the k th prognostic task:

$$AF_k = \frac{1}{k-1} \sum_{j=1}^{k-1} f_j^k \quad (9)$$

where $f_j^k = e_{k,j} - \frac{1}{k-1} \sum_{l=1}^{k-1} e_{l,j}$, $\forall j < k$, denotes the forgetting of the j th task after the model has been continuously trained up to task k ($k > j$). $e_{k,j}$ is the prediction error (e.g. RMSE) assessed on the held-out testing subset of the j th task after continually training the model up to task k .

4.3. Compared methods

To assess the effectiveness of the proposed method, different methods for fault prognosis tasks are implemented for comparison analysis. Note that all these methods apply the same base network for RUL prediction, i.e. the attention-based TCN. The details of them are described as follows.

1. *iid-offline*. Different from the online CL, the CM data used in iid-offline can be shuffled to approach i.i.d and trained offline for multiple epochs with larger batch size. So iid-offline generally shows maximum performance and is regarded as an upper bound of the online CL methods.

2. *iid-online*. In iid-online, the CM data can also be shuffled to be i.i.d, but the model trained with a single epoch and small batch size which is same as other CL methods.

3. *Fine-tuning*. It is a popular baseline in CL study, wherein the base model keeps being trained sequentially task-by-task as the CM data samples continually arrive.

4. *Reservoir*. Reservoir sampling stochastically returns a subset with a constant number of samples that are evenly distributed in an infinite data stream. It is a strong baseline that can be suitable to manage the replay memory in a task-free setting.

5. *Proposed CN-DPM based methods*. Mainly based on the CN-DPM framework, the regression model with GRU networks (CGRU-DPM) is compared with that uses the ATCN (CATCN-DPM) in terms of prediction accuracy. Besides, the performance of using pre-trained model is also compared.

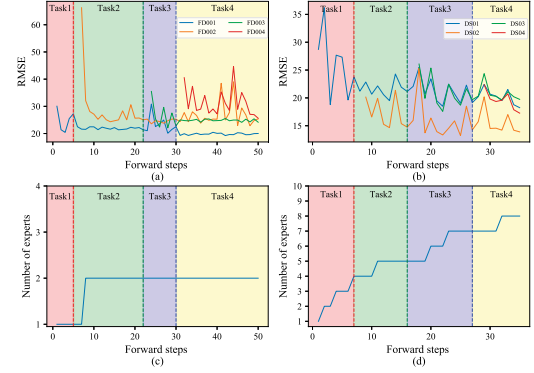


Fig. 6. (a) Testing tasks of RUL prediction results on CMAPSS tasks; (b) Testing tasks of RUL prediction results on N-CMAPSS tasks; (c) The expert growth during the learning process of CMAPSS tasks; (d) The expert growth during the learning process of N-CMAPSS tasks.

4.4. Experimental results and analysis

4.4.1. RUL prediction results of the online task-free CL

The testing RUL prediction results of both two cases obtained by the proposed CL method are shown in Fig. 6, which displays the variation of the prediction accuracy and the growth of the experts over the process of the online CL. As presented in Figs. 6(a) and 6(b), the model performs poorly at the beginning of the training with much higher RMSEs, then the prediction performance shows an upward trend as the learning model is updated continually with more online data samples. Correspondingly, Figs. 6(c) and 6(d) show the expert growth of two cases during the learning process. In detail, for the case of the CMAPSS, the capacity of the model continues to grow as it continues learning new tasks, at first only one expert works on task 1, and then the second expert is created after task 2 comes. Due to different operation conditions, task FD001 has a strong correlation with FD003, and a similar relation exists between task FD002 and FD004. Therefore, the model treats FD001 and FD003 as the same class, and so are the tasks FD002 and FD004, which is consistent with the actual situation, so it can effectively identify the task boundary. For the more realistic and complex case N-CMAPSS, 8 experts have been adaptively created to process this case, where we can find it easier to generate new experts in the early stages of online CL. Other than the models with fixed capacity, the proposed model starts small and grows larger as needed, holding the potential to perform better in online task-free settings.

In addition, Fig. 7 presents the detailed results of RUL prediction on life-time of four engine units sampled from different testing tasks in the CMAPSS case. It can be observed that the RUL prediction curve basically shows a declining trend, in which some obvious degradation patterns are well captured by the model despite some local fluctuations. Moreover, the early prediction errors are generally larger than the latter ones, which is well in line with the real industrial scenarios since

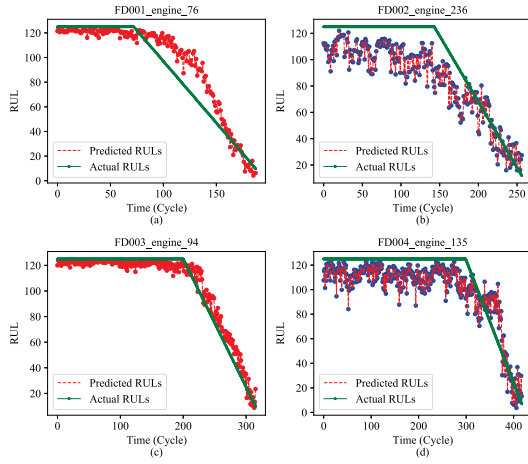


Fig. 7. RUL prediction results of various testing examples in different engines: (a) FD001, engine unit No. 76; (b) FD002, engine unit No. 236; (c) FD003, engine unit No. 94; (d) FD004, engine unit No. 135. Note that the samples with different colors are covered by different experts.

Table 2

Comparison of Prediction performance among different prognostic methods in two cases. **Bold** and underlined texts indicate the best and second performance.

Description	Prognostic Methods	C-MAPSS		N-C-MAPSS	
		RMSE	Score	RMSE	Score
iid baseline	iid-offline	19.57	2981.97	14.21	412.68
	iid-online	26.00	13 740.34	18.69	697.86
CL methods	Fine-tune	33.45	17 149.61	22.32	280.00
	Reservoir	28.57	10 773.72	<u>16.68</u>	349.38
Proposed	CGRU-DPM	25.13	4661.39	17.92	174.14
	CATCN-DPM	<u>23.78</u>	<u>3817.29</u>	17.54	<u>169.19</u>
	CATCN-DPM (pretrained)	24.06	7680.96	17.29	165.40

the degradation behavior of the engine units may not be immediately apparent and can be hard to be captured in the early stage.

4.4.2. Comparison analysis

Several methods with different settings have been implemented to perform comparison analysis, and the average prognostic results of different tasks obtained by these methods on both two cases are reported in Table 2. It shows that the proposed CATCN-DPM method outperforms the other methods under online settings by significant margins in terms of the average RMSE and score values. Compared with the iid-offline setting, our method shows competitive prediction performance and even performs better on N-C-MAPSS tasks in terms of score metric. Besides, among the proposed CN-DPM based methods, the CATCN-DPM method performs better than the CGRU-DPM method. In contrast to RNNs, it also has a simpler and highly parallelizable architecture that holds substantial potential in practical applications. Moreover, it is interesting that the CATCN-DPM with pre-trained model only achieves marginally better prediction results on the N-C-MAPSS case and even performs a little worse in the other case, which illustrates that the pure online CL is capable of learning well from the continuous CM data stream even when the model is trained from scratch.

4.5. Behavior analysis of the experts

To further understand the online continual fault prognosis, this section analyzes the learning behavior of the experts during the online learning process. First, we assess the forgetting of the individual expert through task-wise prediction accuracies at different learning stages. Based on the test prediction results of the responsible expert measured at the end of each task, the forgetting metric is computed to quantify

Table 3

Forgetting of the proposed model for datasets C-MAPSS and N-C-MAPSS.

Methods	C-MAPSS	N-C-MAPSS	Average
Fine-tune	2.96	0.25	1.61
CATCN-DPM	−0.57	−1.81	−1.19

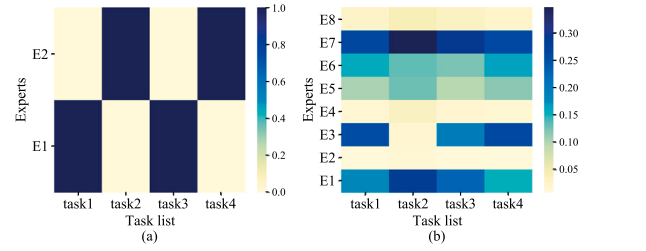


Fig. 8. Sample coverage matrix that displays the statistics of various test samples from different tasks covered by different experts: (a) C-MAPSS dataset; (b) N-C-MAPSS dataset.

how much the method forgets what it learned in the past. Note that higher forgetting value means the method forgets more and vice versa. Table 3 reports the results of the forgetting measure between fine-tune (a natural and popular CL method) and our proposed method. It can be seen that the fine-tune method obtains positive forgetting values for both two cases, i.e. 2.96 and 0.25 respectively, which shows negative knowledge transfer during the continual learning process. In contrast, much lower and negative forgetting values obtained by the proposed method demonstrate that it can effectively overcome the catastrophic forgetting issue, while achieving positive forward knowledge transfer among different tasks.

In addition, to further interpret the learning behavior of the experts on the fault prognosis tasks, Fig. 8 presents the statistics of different test samples covered by different experts by the end of the final training task. As seen in Fig. 8(a), almost all the samples from FD001 and FD003 are covered by the first expert E_1 , while almost all of that from tasks FD002 and FD004 are covered by the second expert E_2 . In contrast, there are 8 different experts dynamically expanded during the whole online continual learning process for N-C-MAPSS, which shows a different learning behavior under a more complex data stream compared with the C-MAPSS data. In detail, it can be observed from Fig. 8(b) that different experts cover various samples from different tasks, among which the coverage states of task 1 (DS01) and task 4 (DS05) are quite similar. We can also see that the sample distribution of task 2 (DS02) covered by various experts is apparently distinguishable when compared with other three tasks, where most samples are covered by expert E_1 and E_7 . Note that task DS02 has two failure modes and other tasks only have a single failure mode, so the learning behavior of various experts is consistent with this discrepancy.

Accordingly, the above analyses show that the proposed model can effectively mitigate the forgetting issue when continually learning different tasks. Moreover, the diversity of expert expansion provides adaptive and efficient coverage of samples across different tasks, which not only validates the effectiveness of the proposed model, but also offers intuitive sights to understand the online continual learning process of the deep model.

5. Conclusion

In this work, an online task-free continual learning-based method is proposed for real-time industrial fault prognosis with the non-stationary CM data stream, where tasks are indistinguishable and the task boundaries are unavailable. By expanding the quantity of experts based on rules under Bayesian nonparametric framework, the proposed method tends to learn as humans do, overcoming the forgetting issue and accumulating knowledge to facilitate the forward prognostic tasks.

Comprehensive experiments demonstrate that the proposed method successfully performs online task-free continual fault prognosis under complex data distribution. Moreover, further analyses show that the proposed model can effectively eliminate the forgetting issue during the CL process, and provide intuitive sights to interpret the learning process of the deep model, which enables the method to be effectively applied in real industries.

Though the proposed method can adaptively expand new experts to handle new tasks, it is essential to acknowledge the potential challenge of generating redundant experts, particularly in scenarios with numerous complex prognostic tasks. In future research, pruning algorithms will be designed to enhance the efficiency of the model. In addition to turbine engines, the proposed method will be applied to more industrial fault prognostic scenarios such as gears and bearings in subsequent research endeavors.

CRedit authorship contribution statement

Chongdang Liu: Writing – original draft, Methodology, Data curation, Conceptualization. **Linxuan Zhang:** Project administration, Funding acquisition. **Yimeng Zheng:** Writing – review & editing. **Zhengyi Jiang:** Data curation, Visualization. **Jinghao Zheng:** Visualization. **Cheng Wu:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

This research was supported by the National Key Technologies Research and Development Program of China under Grant No. 2018YFB1703103.

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